203735: porphyritic metagranodiorite, Lake Lefroy

(Black Flag Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

WIDGIEMOOLTHA (SH 51-14), LAKE LEFROY (3235)
MGA Zone 51, 382112E 6535588N

Sampled on 2 September 2020

This sample was collected from the 218.0 – 219.0 m depth interval of drillcore CD10603, drilled in 2004 by St Ives Gold Mining Company Pty Limited at the Thunderer mine (O'Brien, 2005; Miller et al., 2010). The drillhole is located on the eastern shore of Lake Lefroy, about 17.2 km east of Pilbailey Hill, 14.3 south-southeast of Kambalda, and 11.3 km west of Parker Hill.

Geological context

The unit sampled is a porphyritic metagranodiorite (Fig. 1) assigned to the 2692–2655 Ma Black Flag Group of the Kalgoorlie Terrane (Hall et al., 2022; Smithies et al., 2022). The Black Flag Group consists mainly of felsic metavolcaniclastic and metasedimentary rocks with subordinate felsic and mafic volcanic rocks. The Black Flag Group overlies the 2753–2680 Ma Kalgoorlie Group, is unconformably overlain by the 2664–2655 Ma Merougil Formation (Hall et al., 2022), and in part is intruded by the c. 2680 Ma Golden Mile Dolerite (Rasmussen et al., 2009). A porphyritic hornblende metagranodiorite, sampled from drillhole LD7006 about 1.3 km to the north-northwest, yielded an igneous crystallization age of 2666 ± 7 Ma (GSWA 229913, Lu et al., 2020a). A porphyritic meta-igneous rock, sampled from drillhole NFD003 about 3.0 km to the east, yielded an igneous crystallization age of 2681 ± 5 Ma (GSWA 112110, Nelson, 1995). A porphyritic metagranitic rock, sampled from drillhole CD20297-W1 about 3.2 km to the south-southeast, yielded an igneous crystallization age of 2680 ± 8 Ma (GSWA 203736, Wingate et al., 2022). A porphyritic microgranite, sampled from drillhole LD81635 about 9.7 km to the north-northwest, yielded an igneous crystallization age of 2682 ± 14 Ma (GSWA 230328, Lu et al., 2020b).

Petrographic description

The sample is a medium-grained seriate to sparsely porphyritic metagranodiorite, which contains distinctive diorite and quartz diorite enclaves and subordinate amphibole- and chlorite-rich greenstone xenoliths (Fig. 1). Plagioclase phenocrysts are up to 8 mm in size, within a medium-grained matrix of interlocking plagioclase, amphibole, and minor quartz and biotite. Feldspars are slightly altered to epidote and sericite. Amphibole phenocrysts form mainly subhedral laths up to 2 mm long, partially replaced by actinolite and minor epidote and biotite. Fine-grained biotite is partially altered to chlorite. Accessory minerals include titanite, apatite, zircon, and rare pyrite and iron oxide minerals.

Zircon morphology

The 15 zircons isolated from this sample are colourless to dark brown and mainly subhedral. The crystals are up to 140 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and most crystals are dominated by high-U, metamict zones. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 9–10 and 15–16 October 2020, using SHRIMP-B. Analyses 1.1 to 8.1 (spot numbers 1–8) were obtained during the first session together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.39%. Analyses 9.1 to 11.3 (spot numbers 9–17) were obtained during the second session, together with 16 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seventeen analyses were obtained from 12 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

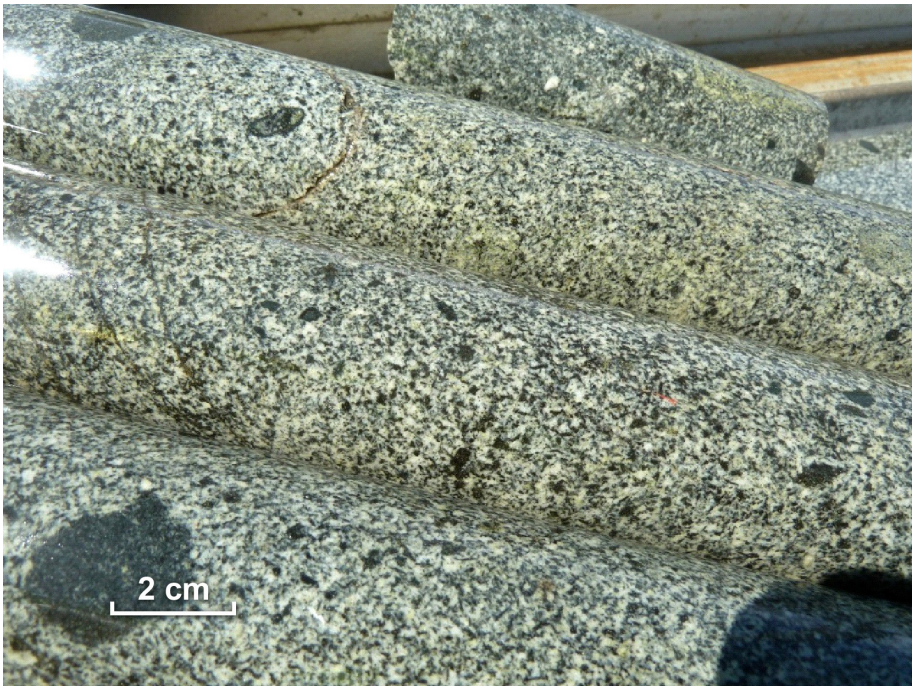

Figure 1. **Drillcore image (wet surface) of sample 203735: porphyritic metagranodiorite, Lake Lefroy**

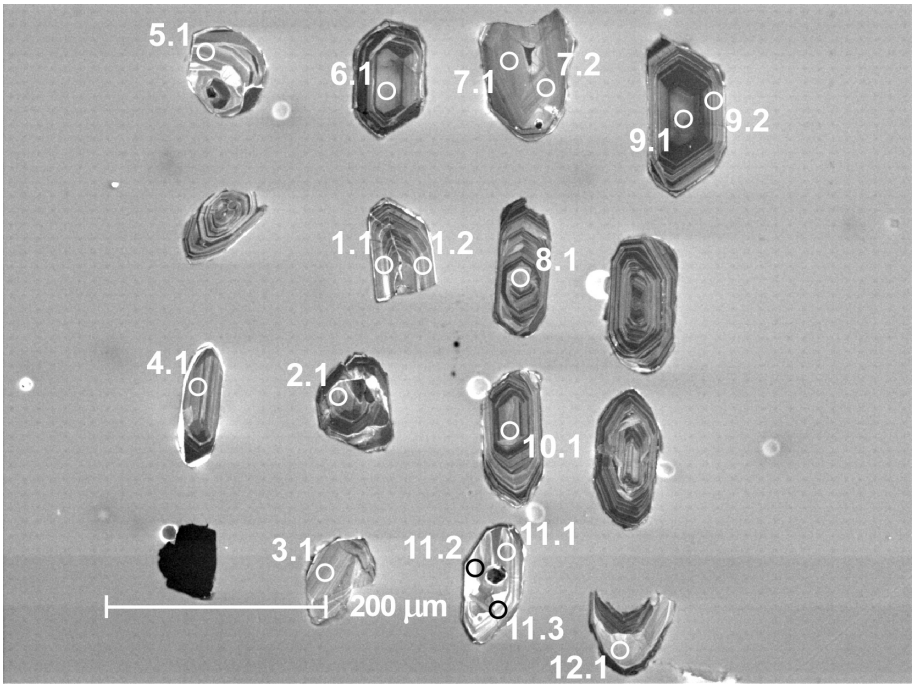

Figure 2. **Cathodoluminescence image of all zircons from sample 203735: porphyritic metagranodiorite, Lake Lefroy. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 203735: porphyritic metagranodiorite, Lake Lefroy

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	8	8.1	147	88	0.60	0.094	2.004	0.028	0.18457	0.00170	2.006	0.028	0.18373	0.00177	2608	30	2687	16	2.9
I	5	5.1	112	22	0.20	0.000	2.027	0.030	0.18381	0.00194	2.027	0.030	0.18381	0.00194	2585	32	2688	17	3.8
I	11	11.1	98	80	0.82	0.056	1.995	0.036	0.18505	0.00204	1.996	0.036	0.18455	0.00210	2618	39	2694	19	2.8
I	7	7.1	134	85	0.64	-0.033	2.000	0.029	0.18481	0.00176	1.999	0.029	0.18510	0.00178	2615	31	2699	16	3.1
I	15	9.2	176	177	1.01	0.848	2.059	0.030	0.19315	0.00162	2.077	0.031	0.18558	0.00222	2534	31	2703	20	6.3
I	17	11.3	76	39	0.51	-0.172	1.912	0.042	0.18411	0.00250	1.909	0.042	0.18564	0.00272	2715	49	2704	24	-0.4
I	10	10.1	209	100	0.48	0.492	1.963	0.026	0.19052	0.00349	1.973	0.027	0.18613	0.00365	2644	29	2708	32	2.4
I	1	1.1	88	42	0.47	-0.320	1.939	0.064	0.18454	0.00199	1.933	0.063	0.18740	0.00226	2688	72	2719	20	1.2
P	4	4.1	125	71	0.57	0.375	2.033	0.029	0.18195	0.00173	2.041	0.029	0.17859	0.00201	2571	31	2640	19	2.6
P	3	3.1	90	41	0.46	0.230	1.908	0.031	0.18082	0.00197	1.913	0.032	0.17876	0.00218	2711	37	2641	20	-2.6
P	12	12.1	82	66	0.81	0.068	1.995	0.039	0.17947	0.00218	1.996	0.039	0.17886	0.00227	2618	43	2642	21	0.9
P	14	7.2	127	71	0.56	0.178	1.972	0.033	0.18125	0.00178	1.976	0.033	0.17967	0.00195	2640	36	2650	18	0.4
P	13	11.2	63	47	0.74	-0.103	1.915	0.045	0.17881	0.00267	1.913	0.045	0.17973	0.00282	2710	53	2650	26	-2.3
P	16	1.2	102	60	0.59	0.115	1.999	0.070	0.18148	0.00201	2.001	0.070	0.18045	0.00214	2613	75	2657	20	1.7
P	6	6.1	197	138	0.70	0.023	1.940	0.024	0.18078	0.00149	1.941	0.024	0.18057	0.00150	2679	27	2658	14	-0.8
P	2	2.1	189	147	0.78	0.195	2.092	0.042	0.18348	0.00148	2.096	0.042	0.18174	0.00159	2514	42	2669	15	5.8
D	9	9.1	240	413	1.72	0.384	2.707	0.089	0.19092	0.00143	2.718	0.090	0.18749	0.00169	2020	57	2720	15	25.8

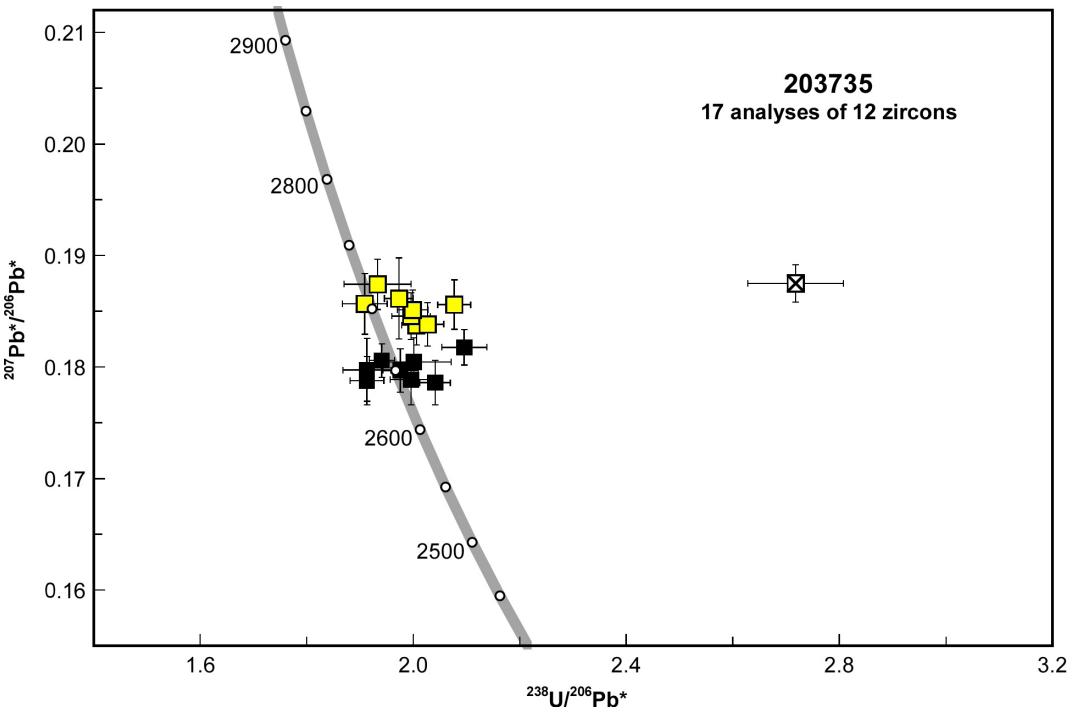

Figure 3. **U–Pb analytical data for sample 203735: porphyritic metagranodiorite, Lake Lefroy. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed square indicates Group D (discordance >7%)**

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). One analysis is >7% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and is considered not to be geologically significant. The remaining 16 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises eight analyses of seven zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2698 ± 13 Ma (MSWD = 0.33).

Group P comprises eight analyses of eight zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2669–2640 Ma, and a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2653 ± 12 Ma (MSWD = 0.36). Three analyses in Group P yield $^{207}Pb^*/^{206}Pb^*$ dates that are significantly younger than additional analyses (assigned to Group I) in the same zircons, and three more analyses yield dates that are even younger.

The date of 2698 ± 13 Ma for the eight analyses in Group I is tentatively interpreted as the age of igneous crystallization. The dates of 2669–2640 Ma for the eight analyses in Group P are interpreted to reflect loss of radiogenic Pb, because some dates are significantly younger than additional analyses in the same zircons. It is possible that one or two oldest analyses included in Group P should be assigned to Group I, in which case the interpreted crystallization age would be younger by up to 11 Ma.

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Recommended reference for this publication

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